

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

</tool_call> reasoning process here

The de Broglie wavelength of a particle is given by the formula :

$\lambda = h / p$

where :

- h is Planck's constant = 6.626×10^{-34} J·s (or 6.626×10^{-27} erg·s),
- p is the momentum of the particle .

For a non-relativistic electron , kinetic energy $K = (1/2)mv^2$, and momentum $p = mv$.

So , we can write :

$p = \sqrt{2mK}$

Therefore , de Broglie wavelength :

$\lambda = h / \sqrt{2mK}$

First , convert kinetic energy to joules :

$1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$

So $K = 1.602 \times 10^{-19}\text{ J}$

Mass of electron $m = 9.109 \times 10^{-31}\text{ kg}$

Now plug into the formula :

$\lambda = (6.626 \times 10^{-34}) / \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}$

Compute inside square root :

$2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19} = 2.92 \times 10^{-49}$

$\sqrt{2.92 \times 10^{-49}} \approx 5.40 \times 10^{-25}$

Then :

$\lambda = 6.626 \times 10^{-34} / 5.40 \times 10^{-25} \approx 1.227 \times 10^{-9}\text{ m}$

Convert to cm : $1.227 \times 10^{-9}\text{ m} = 1.227 \times 10^{-7}\text{ cm}$

Now , check options in cm :

- A. $2.48 \times 10^{-8}\text{ cm} \approx 2.48 \times 10^{-8}$
- B. $5.92 \times 10^7\text{ cm}$ $\approx$ too large
- C. $8.66 \times 10^{-6}\text{ cm}$
- D. $3.14 \times 10^{-5}\text{ cm}$
- E. $9.81 \times 10^{-4}\text{ cm}$
- F. $5.27 \times 10^{-9}\text{ m} \approx 5.27 \times 10^{-10}\text{ m} = 5.27 \times 10^{-8}\text{ cm}$ $\approx$ not matching
- G. $1.23 \times 10^{-7}\text{ cm} \approx$ this is $1.23 \times 10^{-7}\text{ cm} \approx$ which is $1.23 \times 10^{-6}\text{ m} \approx 1.23 \times 10^{-9}\text{ m}$
- Wait $\rightarrow$ let's verify conversion :

$1.227 \times 10^{-9}\text{ m} = 1.227 \times 10^{-7}\text{ cm}$
Yes because $1\text{ m} = 100\text{ cm} \rightarrow 1 \times 10^{-9}\text{ m} = 1 \times 10^{-7}\text{ cm} \rightarrow$ so $1.227 \times 10^{-9}\text{ m} = 1.227 \times 10^{-7}\text{ cm}$

So $\lambda \approx 1.23 \times 10^{-7}\text{ cm}$

Option G is : $1.23 \times 10^{-7}\text{ cm} \approx$ matches

Convert option G to m : $1.23 \times 10^{-7}\text{ cm} = 1.23 \times 10^{-9}\text{ m} \approx$ known as $1.23 \times 10^{-9}\text{ m}$?

Wait $10^{-7}\text{ cm} = 10^{-9}\text{ m} \approx$ so $1.23 \times 10^{-7}\text{ cm} = 1.23 \times 10^{-9}\text{ m} \approx$ erg , yes

Option H is $4.14 \times 10^{-10}\text{ m} = 4.14 \times 10^{-8}\text{ cm} \approx$ not same

So wavelength = $\sim 1.23 \times 10^{-9}\text{ m}$